Full Report

sales_large.csv · 8,000 rows · 6 columns · March 19, 2026

This report provides a structured analysis of **sales_large.csv**, covering 8,000 records across 6 columns. It examines distribution patterns, and trends over time, informed by the context and goals provided. Key findings, recommendations, and assumptions are outlined in the sections below.

DATASETS

customers_large.csv **sales_large.csv**

RELATIONSHIPS

customers_large.csv · customer_id → sales_large.csv · customer_id

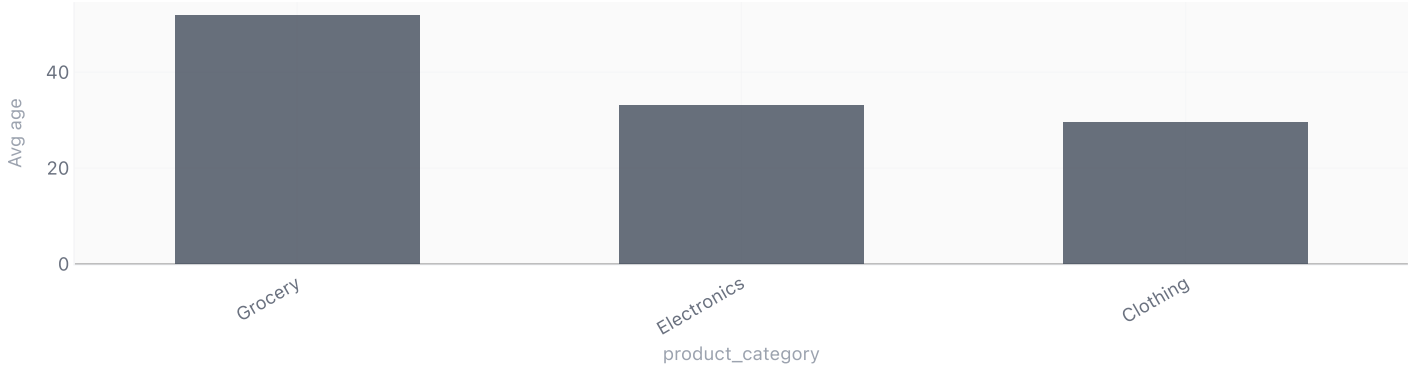
Column 'customer_id' appears in both files (string), 100% value overlap

Matched	1,200	100.0% of smaller dataset
Only in customers_large.csv	0	no match in sales_large.csv
Only in sales_large.csv	0	no match in customers_large.csv

age by product_category

After joining customers_large.csv and sales_large.csv on 'customer_id', age (from customers_large.csv) broken down by product_category (from sales_large.csv).

age by product_category



- Grocery: avg 51.89 (+33% vs overall)
- Electronics: avg 33.04 (-16% vs overall)
- Clothing: avg 29.41 (-25% vs overall)

PROJECT BACKGROUND

The dataset under consideration, titled "sales_large.csv," encompasses a comprehensive collection of sales transactions, comprising 8,000 records across six columns. It captures key metrics related to sales performance, specifically focusing on the monetary amount of sales. The dataset includes dimensions such as product category and sales channel, which provide insights into the various segments of the market. The presence of a datetime field suggests that the data spans a specific time period, allowing for temporal analysis of sales trends and patterns. The primary analytical objective of this report is to identify top-performing product categories and sales channels to inform strategies aimed at increasing overall revenue. By understanding which segments contribute most significantly to sales, stakeholders can make data-driven decisions to enhance performance and optimize resource allocation. This analysis is crucial for developing targeted marketing strategies and improving sales tactics, ultimately driving revenue growth. To achieve these goals, the analysis will explore various dimensions and measures within the dataset. Key patterns such as trends over time, differences in performance across product categories, and correlations between sales channels will be examined. This multifaceted approach will provide a clearer picture of sales dynamics, enabling the identification of opportunities for improvement and growth within the organization.

EXECUTIVE SUMMARY

- Interestingly, the average sales amount for Electronics is significantly higher at 268.9 compared to Grocery at 120.29, suggesting that focusing marketing efforts on Electronics could yield higher revenue growth.
- What stands out here is the performance disparity between sales channels; Online sales average 243.43 while In-Store sales are at 164.06. This indicates a potential opportunity to enhance the in-store experience or promotions to boost sales in that channel.
- Notably, there is a slight downward trend in overall sales amount, decreasing by 0.4% from the first half to the second half of the year. This suggests a need to investigate the factors contributing to this decline and implement strategies to reverse it.
- The standard deviation of 116.579 in sales amount indicates a wide variability in transaction sizes, which could imply that while some customers are making large purchases, others are buying much less. Targeting high-value customers with tailored promotions could help increase overall revenue.
- With 2881 transactions in the Electronics category, which is the top performer, it may be beneficial to analyze the specific products driving these sales to identify potential areas for expansion or additional marketing focus.
- The spread percentage of 75.4% between the top and bottom product categories suggests a significant opportunity to elevate the performance of lower-performing categories like Grocery. Investigating customer preferences in this category could uncover actionable insights.
- Given that the mean sales amount for Online transactions is 40.2% higher than In-Store, it may be worth exploring the reasons behind this preference and leveraging successful online strategies to enhance in-store sales.

DATA STRUCTURE

COLUMN	TYPE	FLAGS	NULLS	UNIQUE	MEAN	MIN	MAX
sale_id	string	id like	0.0%	8,000	·	·	·
customer_id	string	id like	0.0%	1,200	·	·	·
date	datetime	·	0.0%	90	·	·	·
product_category	string	low cardinality	0.0%	3	·	·	·
amount	numeric	·	0.0%	7,139	197.1963	9.8	580.13
channel	string	low cardinality	0.0%	2	·	·	·

INSIGHTS DEEP-DIVE

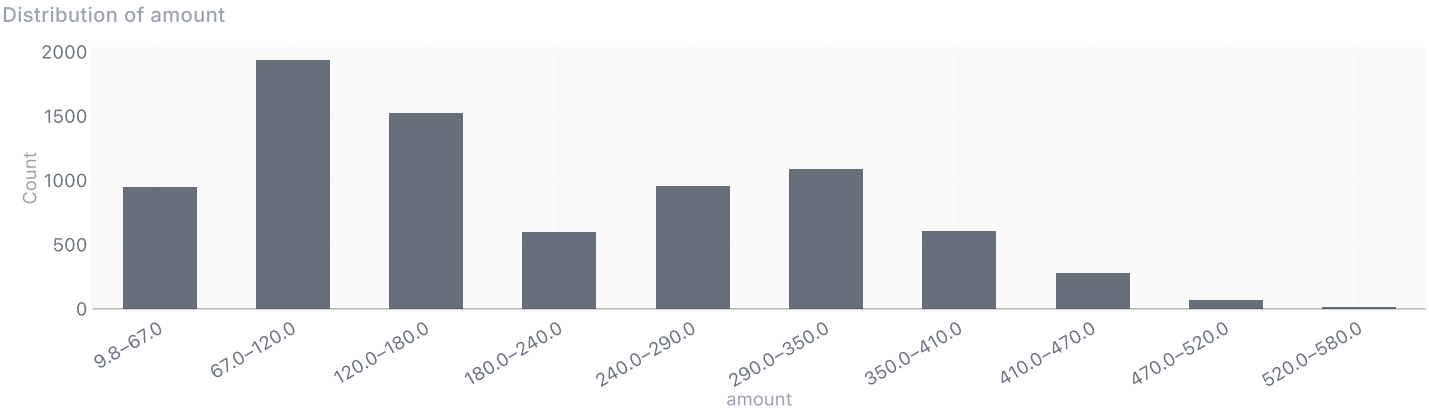
Dataset Overview

'sales_large.csv' — 8,000 rows × 6 columns.

- Measures (1): amount
- Dimensions (2): product_category, channel
- Time dimensions (1): date
- Excluded from analysis (2): sale_id (id_like), customer_id (id_like)

Distribution: amount

Shape and spread of 'amount' across 8,000 non-null records.

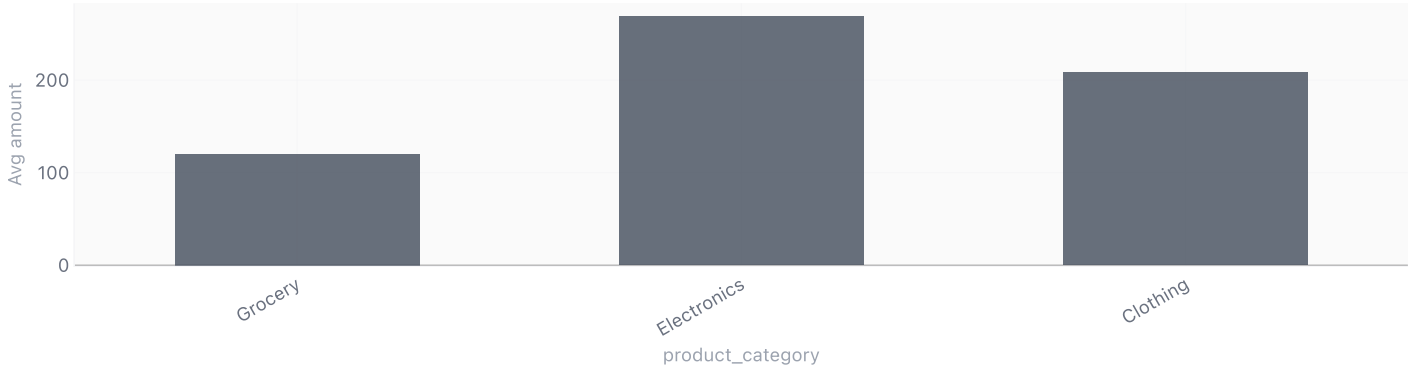


- Range: 9.8 to 580.13 (span of 570.33)
- Central tendency: mean 197.20, median 156.60
- Spread: standard deviation 116.58
- Distribution is roughly symmetric — mean and median are close (40.59 apart).
- Moderate variability (CV = 0.59) — meaningful spread around the average.

product_category Breakdown

'product_category' has 3 distinct value(s). Top 3 groups by record count, with mean amount:

amount by product_category

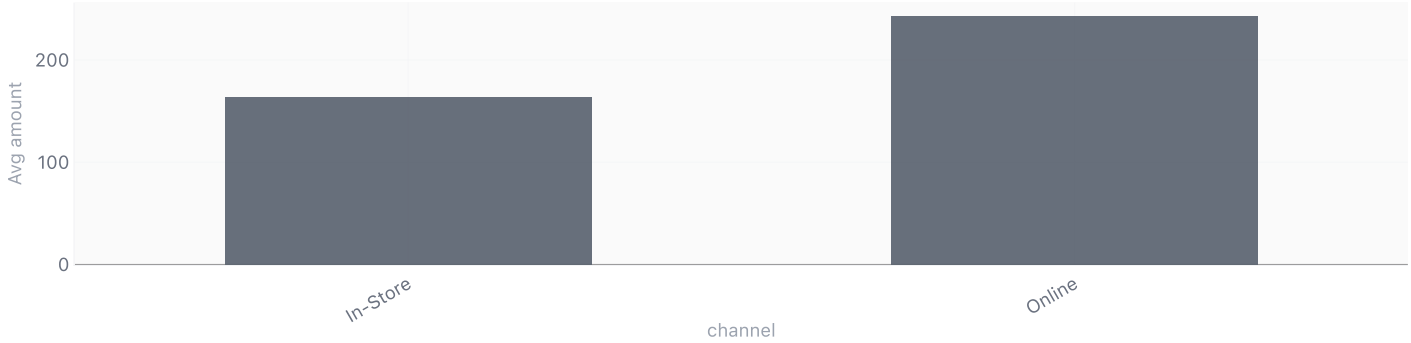


- Grocery (2,998 records) — avg amount: 120.29
- Electronics (2,881 records) — avg amount: 268.9
- Clothing (2,121 records) — avg amount: 208.51

channel Breakdown

'channel' has 2 distinct value(s). Top 2 groups by record count, with mean amount:

amount by channel

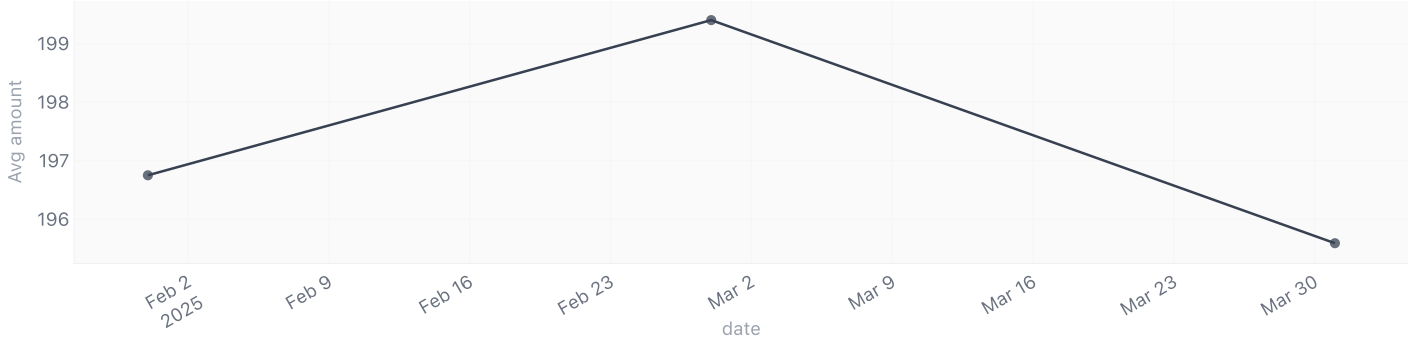


- In-Store (4,660 records) — avg amount: 164.06
- Online (3,340 records) — avg amount: 243.43

amount over Time

How 'amount' changes across time periods in 'date'.

amount over Time (Monthly avg)



- 3 time periods shown.
- Peak: 2025-02-28 00:00:00 at 199.40.
- Trough: 2025-03-31 00:00:00 at 195.59.
- Second half of the period averages 0.4% higher than the first half.

Only 3 time periods available — trend direction may not be reliable with this few data points.

RECOMMENDATIONS

The following recommendations are grounded in the patterns identified above. Each recommendation is paired with the observation that motivates it.

01 Focus Marketing on Electronics Category

- The average sales amount for Electronics is significantly higher at 268.9 compared to Grocery at 120.29, indicating a potential for greater revenue growth.
- Allocate more marketing resources towards Electronics to capitalize on its higher sales potential.

02 Enhance In-Store Sales Experience

- Online sales average 243.43 while In-Store sales are at 164.06, highlighting a performance disparity between sales channels.
- Implement targeted promotions and improve the in-store shopping experience to boost In-Store sales.

03 Investigate Sales Decline Factors

- There is a slight downward trend in overall sales amount, decreasing by 0.4% from the first half to the second half of the year.
- Conduct a thorough analysis to identify factors contributing to the sales decline and develop strategies to reverse this trend.

04 Target High-Value Customers with Promotions

- The standard deviation of 116.579 in sales amount indicates wide variability in transaction sizes, suggesting some customers make large purchases while others do not.
- Create tailored promotions aimed at high-value customers to increase overall revenue.

05 Analyze Top-Selling Electronics Products

- With 2881 transactions in the Electronics category, which is the top performer, there is an opportunity to identify key products driving these sales.

- Analyze the specific products within the Electronics category to identify areas for expansion or additional marketing focus.

06 Explore Online Strategies for In-Store Sales

- The mean sales amount for Online transactions is 40.2% higher than In-Store, suggesting a preference for online shopping.
- Investigate successful online sales strategies and adapt them to enhance In-Store sales performance.

ASSUMPTIONS

- Assume that the sales data accurately reflects all transactions made during the period covered by the dataset.
- Assume that the excluded columns 'sale_id' and 'customer_id' do not contain critical information necessary for the analysis.
- Assume that the datetime information is consistently formatted and correctly represents the time of each sale.

LIMITATIONS

- This analysis cannot determine the reasons behind fluctuations in sales since customer demographics are not included.
- The dataset does not provide information on the geographical distribution of sales, limiting regional analysis.
- No insights can be drawn regarding customer behavior or preferences due to the exclusion of customer-related data.

CONCLUSION

The analysis of the sales_large.csv dataset reveals several critical insights that can guide our strategy to push top performers and increase overall revenue. Notably, the average sales amount for Electronics stands at 268.9, significantly outpacing Grocery's 120.29, indicating that a targeted marketing push in the Electronics category could drive substantial revenue growth. Additionally, the stark performance disparity between sales channels—where Online sales average 243.43 compared to In-Store sales at 164.06—highlights an opportunity to enhance the in-store experience or promotions to boost sales in that channel. The slight 0.4% decline in overall sales from the first half to the second half of the year suggests a need for further investigation into the underlying factors contributing to this trend, while the wide variability in transaction sizes, evidenced by a standard deviation of 116.579, points to the potential for tailored promotions aimed at high-value customers. Moving forward, it would be prudent to analyze the specific products driving the 2881 transactions in Electronics and explore customer preferences in lower-performing categories like Grocery to uncover actionable insights that can elevate overall performance.

KEY FINDINGS

- Interestingly, the average sales amount for Electronics is significantly higher at 268.9 compared to Grocery at 120.29, suggesting that focusing marketing efforts on Electronics could yield higher revenue growth.
- What stands out here is the performance disparity between sales channels; Online sales average 243.43 while In-Store sales are at 164.06. This indicates a potential opportunity to enhance the in-store experience or promotions to boost sales in that channel.
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DASHBOARD

8,000
TOTAL ROWS

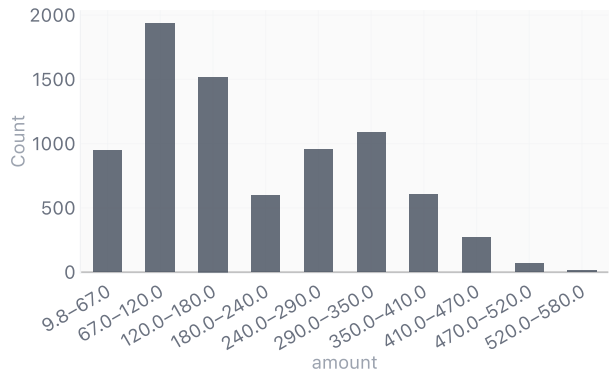
6
COLUMNS

0.0%
AVG NULL RATE

5
INSIGHTS

Distribution: amount

Distribution of amount



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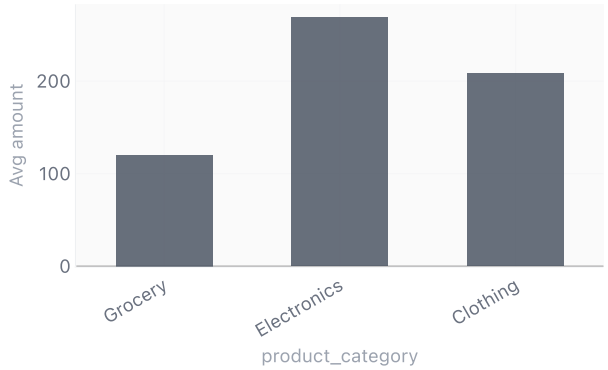

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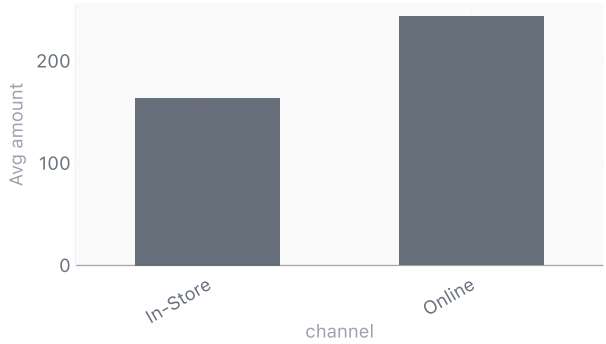

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